STANDARD 2
Stage 6

Algebra (Std 2), A4 Types of Relationships (Y12)

Non-Linear: Inverse and Other Problems (Std 2)

Teacher:

Exam Equivalent Time: 69 minutes *(based on allocation of 1.5 minutes per mark)*

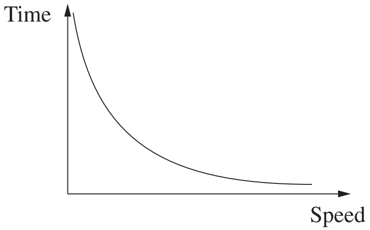
Questions

1. Algebra, STD2 A1 2009 HSC 16 MC

The time for a car to travel a certain distance varies inversely with its speed.

Which of the following graphs shows this relationship?

(A)



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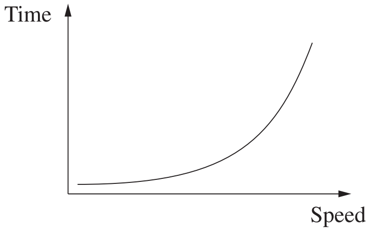

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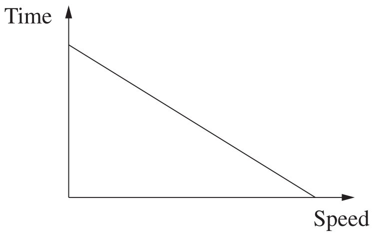

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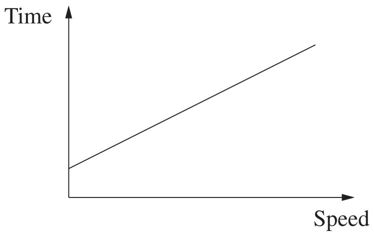

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2. Algebra, STD2 A4 2010 HSC 13 MC

The number of hours that it takes for a block of ice to melt varies inversely with the temperature. At 30°C it takes 8 hours for a block of ice to melt.

How long will it take the same size block of ice to melt at 12°C?

A. 3.2 hours

B. 20 hours

C. 26 hours

D. 45 hours

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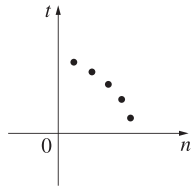
3. Algebra, STD2 A4 2004 HSC 21 MC

The time (t) taken to clean a house varies inversely with the number (n) of people cleaning the house.

Which graph represents this relationship?

(A)



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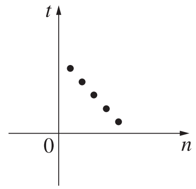

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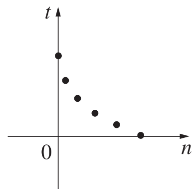
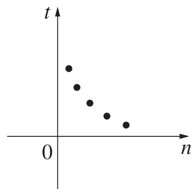


(D)



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4. Algebra, STD2 A4 2007 HSC 15 MC

If pressure (p) varies inversely with volume (V), which formula correctly expresses p in terms of V and k, where k is a constant?

A. $p = \dfrac{k}{V}$

B. $p = \dfrac{V}{k}$

C. $p = kV$

D. $p = k + V$

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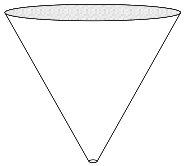
5. Algebra, STD1 A3 2019 HSC 9 MC

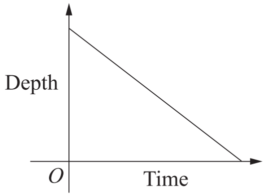
The container shown is initially full of water.



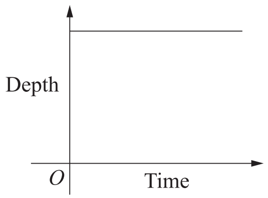
Water leaks out of the bottom of the container at a constant rate.

Which graph best shows the depth of water in the container as time varies?

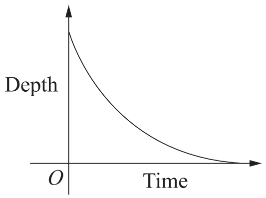
A.



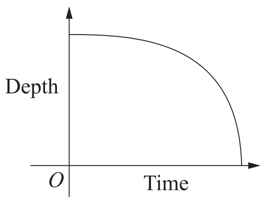
B.



C.



D.



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6. Algebra, STD2 A4 2021 HSC 13 MC

The time taken to clean a warehouse varies inversely with the number of cleaners employed.

It takes 8 cleaners 60 hours to clean a warehouse.

Working at the same rate, how many hours would it take 10 cleaners to clean the same warehouse.

A. 45

B. 48

C. 62

D. 75

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7. Algebra, STD2 A4 2024 HSC 9 MC

The time taken to paint a school varies inversely with the number of painters completing the task.

It takes 6 painters a total of 20 days to paint a school.

How many days would it take 15 painters to paint the same school?

A. 4.5

B. 8

C. 15.5

D. 50

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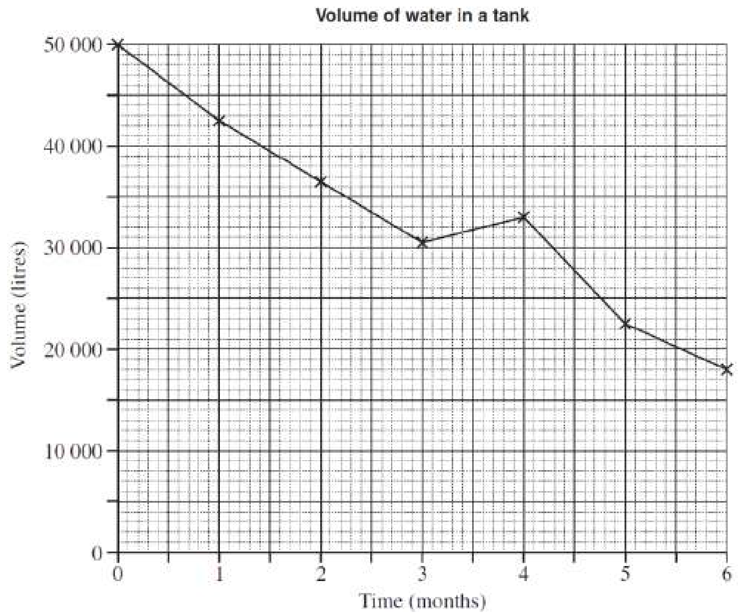
8. Algebra, STD2 A2 2009 HSC 13 MC

The volume of water in a tank changes over six months, as shown in the graph.



Consider the overall decrease in the volume of water.

What is the average percentage decrease in the volume of water per month over this time, to the nearest percent?

A. 6%

B. 11%

C. 32%

D. 64%

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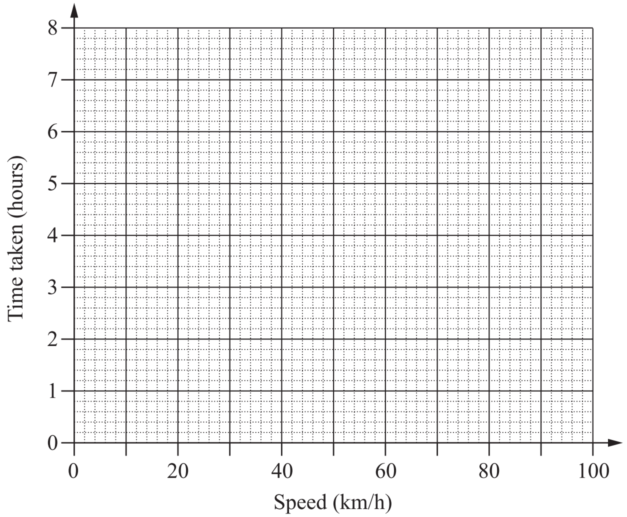
9. Algebra, STD2 A4 2019 HSC 33

The time taken for a car to travel between two towns at a constant speed varies inversely with its speed.

It takes 1.5 hours for the car to travel between the two towns at a constant speed of 80 km/h.

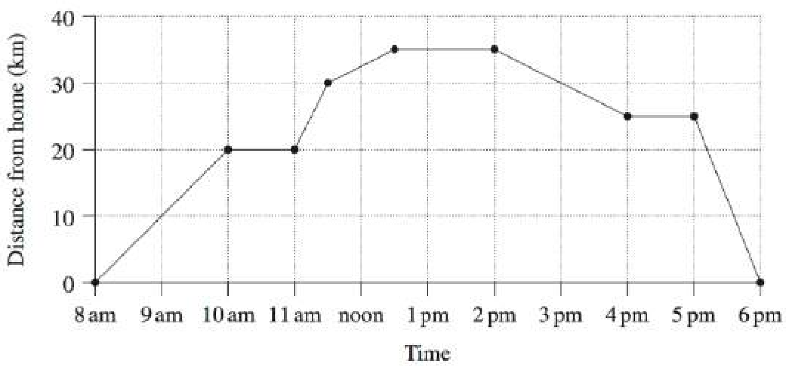
a. Calculate the distance between the two towns. ***(1 mark)***

b. By first plotting four points, draw the curve that shows the time taken to travel between the two towns at different constant speeds. ***(3 marks)***



10. Algebra, STD2 A4 2013 HSC 27a

Lucy went for a bike ride. She left home at 8 am and arrived back at home at 6 pm. A graph representing her journey is shown.



i. What was the total distance that she rode during the day? ***(1 mark)***

ii. How much time did Lucy spend riding her bike during the day? ***(1 mark)***

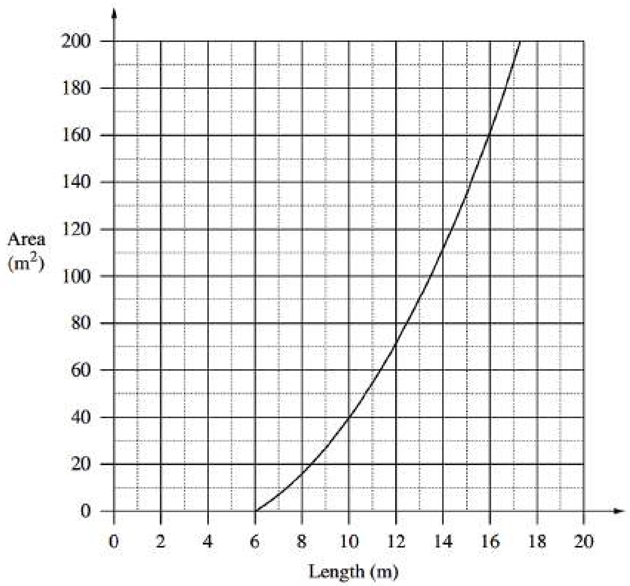
11. Algebra, STD2 A4 2007 HSC 27a

A rectangular playing surface is to be constructed so that the length is 6 metres more than the width.

i. Give an example of a length and width that would be possible for this playing surface. ***(1 mark)***

ii. Write an equation for the area (A) of the playing surface in terms of its length (l). ***(1 mark)***

A graph comparing the area of the playing surface to its length is shown.



iii. Why are lengths of 0 metres to 6 metres impossible? ***(1 mark)***

iv. What would be the dimensions of the playing surface if it had an area of 135 m²? ***(2 marks)***

Company A constructs playing surfaces.

Company *A* charges

Size of playing surface	*Charges*
Up to and including 150 m^2	\$50 000
Greater than 150 m^2	\$50 000 plus a rate of \$300 per square metre for the area in excess of 150 m^2

v. Draw a graph to represent the cost of using Company A to construct all playing surface sizes up to and including 200 m².

Use the horizontal axis to represent the area and the vertical axis to represent the cost. ***(2 marks)***

vi. Company B charges a rate of \$360 per square metre regardless of size.

Which company would charge less to construct a playing surface with an area of 135 m²

Justify your answer with suitable calculations. ***(1 mark)***

12. Algebra, STD2 A4 2011 HSC 28a

The air pressure, P, in a bubble varies inversely with the volume, V, of the bubble.

i. Write an equation relating P, V and a, where a is a constant. ***(1 mark)***

ii. It is known that $P = 3$ when $V = 2$.

By finding the value of the constant, a, find the value of P when $V = 4$. ***(2 marks)***

iii. Sketch a graph to show how P varies for different values of V.

Use the horizontal axis to represent volume and the vertical axis to represent air pressure. ***(2 marks)***

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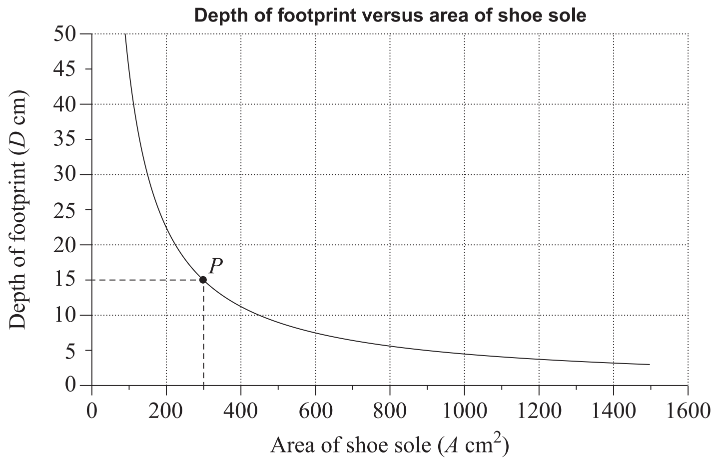
13. Algebra, STD2 A4 2018 HSC 29c

When people walk in snow, the depth (D cm) of each footprint depends on both the area (A cm²) of the shoe sole and the weight of the person. The graph shows the relationship between the area of the shoe sole and the depth of the footprint in snow, for a group of people of the same weight.



i. The graph is a hyperbola because D is inversely proportional to A. The point P lies on the hyperbola.

Find the equation relating D and A. ***(2 marks)***

ii. A man from this group walks in snow and the depth of his footprint is 4 cm.

Use your equation from part (i) to calculate the area of his shoe sole. ***(1 mark)***

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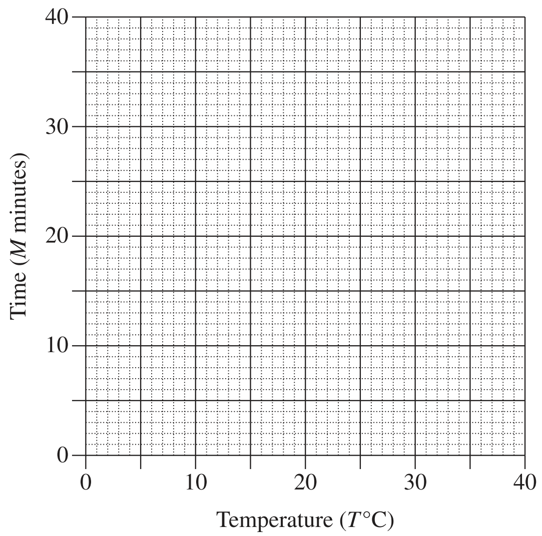
14. Algebra, STD2 A4 2022 HSC 24

A student believes that the time it takes for an ice cube to melt (M minutes) varies inversely with the room temperature $(T°\text{C})$. The student observes that at a room temperature of $15°\text{C}$ it takes 12 minutes for an ice cube to melt.

a. Find the equation relating M and T. ***(2 marks)***

b. By first completing this table of values, graph the relationship between temperature and time from $T = 5°C$ to $T = 30°\text{C}$. ***(2 marks)***

T	5	15	30
M			



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15. Algebra, STD2 A4 SM-Bank 10

The number of trees that can be planted along the fence line of a paddock varies inversely with the distance between each tree.

There will be 108 trees if the distance between them is 5 metres.

i. How many trees can be planted if the distance between them is 6 metres? ***(2 marks)***

ii. What is the distance between the trees if 120 trees are planted. ***(1 mark)***

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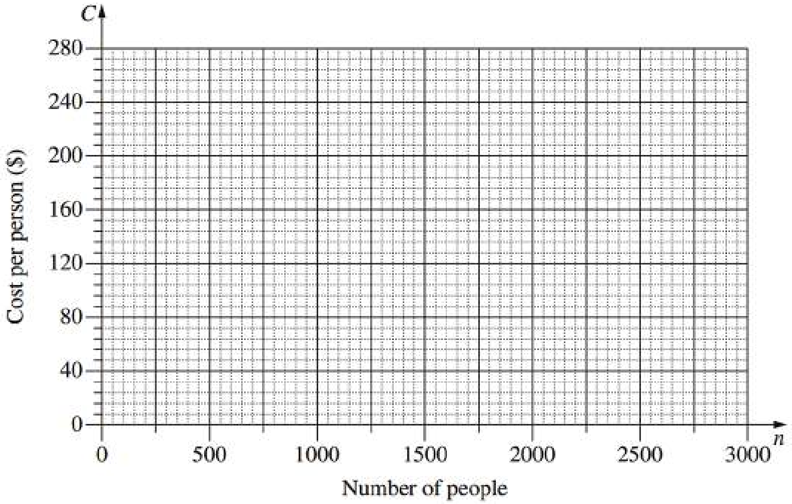
16. Algebra, STD2 A4 2014 HSC 29a

The cost of hiring an open space for a music festival is \$120 000. The cost will be shared equally by the people attending the festival, so that C (in dollars) is the cost per person when n people attend the festival.

i. Complete the table below by filling in the THREE missing values. ***(1 mark)***

Number of people (n)	500	1000	1500	2000	2500	3000
Cost per person (C)				60	48	40

ii. Using the values from the table, draw the graph showing the relationship between n and C. ***(2 marks)***



iii. What equation represents the relationship between n and C? ***(1 mark)***

iv. Give ONE limitation of this equation in relation to this context. ***(1 mark)***

v. Is it possible for the cost per person to be \$94? Support your answer with appropriate calculations. ***(1 mark)***

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17. Algebra, STD2 A4 2004 HSC 28a

A health rating, R, is calculated by dividing a person's weight, w, in kilograms by the square of the person's height, h, in metres.

i. Fred is 150 cm and weighs 72 kg. Calculate Fred's health rating. ***(1 mark)***

ii. Over several years, Fred expects to grow 10 cm taller. By this time he wants his health rating to be 25. How much weight should he gain or lose to achieve his aim? Justify your answer with mathematical calculations. ***(2 marks)***

STANDARD 2

Stage 6

Algebra (Std 2), A4 Types of Relationships (Y12)

Non-Linear: Inverse and Other Problems (Std 2)

Teacher:

Exam Equivalent Time: 69 minutes *(based on allocation of 1.5 minutes per mark)*

Worked Solutions

1. Algebra, STD2 A1 2009 HSC 16 MC

$$T \propto \frac{1}{S}$$

$$T = \frac{k}{S}$$

By elimination:

As S$\uparrow$, T$\downarrow$ $\Rightarrow$ cannot be B or D

C is incorrect because it graphs a linear relationship

$\Rightarrow A$

♦ *Mean mark 38%*

2. Algebra, STD2 A4 2010 HSC 13 MC

♦ *Mean mark 50%*

Time to melt $(T) \propto \dfrac{1}{\text{Temp}} \Rightarrow T = \dfrac{k}{\text{Temp}}$

When $T = 8$, Temp $= 30$

$$8 = \frac{k}{30}$$

$$k = 240$$

Find T when Temp $= 12$:

$$T = \frac{240}{12}$$

$$= 20 \text{ hours}$$

$\Rightarrow B$

3. Algebra, STD2 A4 2004 HSC 21 MC

$t \propto \dfrac{1}{n} \Rightarrow t = \dfrac{k}{n}$

The graph is a hyperbola that sees t decrease as n increases (eliminate A and B).

Also, t cannot be zero (eliminate C).

$\Rightarrow D$

4. Algebra, STD2 A4 2007 HSC 15 MC

$$p \propto \frac{1}{V}$$

$$p = \frac{k}{V}$$

$\Rightarrow A$

5. Algebra, STD1 A3 2019 HSC 9 MC

Depth will decrease slowly at first and accelerate.

$\Rightarrow D$

6. Algebra, STD2 A4 2021 HSC 13 MC

Time to clean $(T) \propto \dfrac{1}{\text{Number of cleaners (C)}}$

♦ *Mean mark 41%.*

$$T = \frac{k}{C}$$

When $T = 60, C = 8$

$$60 = \frac{k}{8}$$

$$k = 480$$

Find T when $C = 10$:

$$T = \frac{480}{10}$$

$$= 48 \text{ hours}$$

$\Rightarrow B$

7. Algebra, STD2 A4 2024 HSC 9 MC

$$T \propto \frac{1}{N} \Rightarrow T = \frac{k}{N}$$

Find k given $T = 20$ when $N = 6$:

$$20 = \frac{k}{6} \Rightarrow k = 120$$

$$\therefore T = \frac{120}{N}$$

Find T if $N = 15$:

$$T = \frac{120}{15} = 8 \text{ days}$$

$\Rightarrow B$

8. Algebra, STD2 A2 2009 HSC 13 MC

$$\text{Initial Volume} = 50\ 000 \text{ L}$$

$$\text{Final volume} = 18\ 000 \text{ L}$$

$$\text{Decrease} = 50\ 000 - 18\ 000$$

$$= 32\ 000 \text{ L} \quad \text{(over 6 months)}$$

♦ *Mean mark 48%*

COMMENT: Remember that % decrease requires the decrease in volume to be divided by the *original* volume (50,000L).

$$\text{Loss per month} = \frac{32\ 000}{6}$$

$$= 5333.33\ldots \text{ L per month}$$

$$\%\text{ loss per month} = \frac{5333.33\ldots}{50\ 000}$$

$$= 10.666\ldots\%$$

$\Rightarrow B$

9. Algebra, STD2 A4 2019 HSC 33

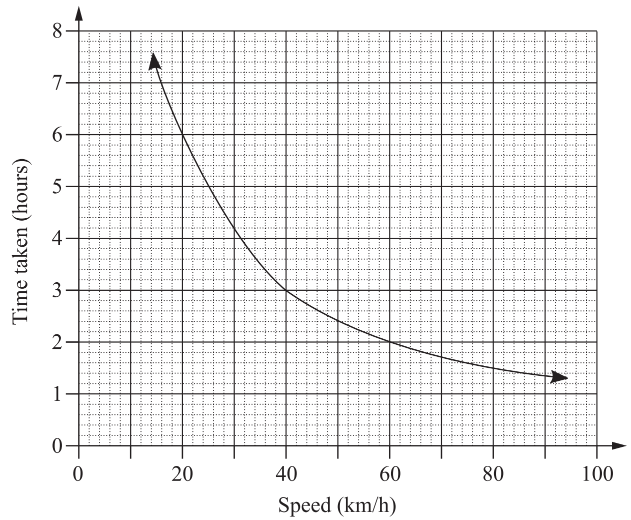
a. $D = S \times T$

$$= 80 \times 1.5$$

$$= 120 \text{ km}$$

b.

s	20	40	60	80
t	6	3	2	1.5



10. Algebra, STD2 A4 2013 HSC 27a

i. $\text{Total distance} = 35 + 35$

$= 70\ \text{km}$

ii. $\text{Time away from home} = 10\ \text{hours}$

$\text{Time resting} = 1 + 1.5 + 1$

$= 3.5\ \text{hours}$

$\therefore\ \text{Time riding} = 10 – 3.5$

$= 6.5\ \text{hours}$

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11. Algebra, STD2 A4 2007 HSC 27a

i. One possibility is a length of 10 m, and a width of 4 m (among many possibilities).

ii. $\text{Length} = l\ \text{m}$

$\text{Width} = (l – 6)\ \text{m}$

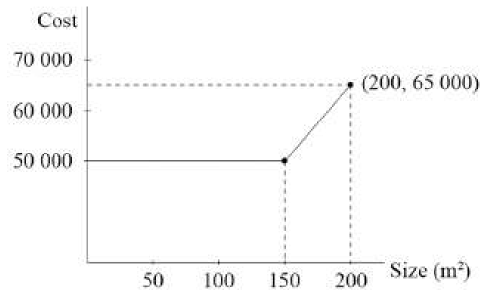
$\therefore\ A = l(l – 6)$

iii. Given the length must be 6m more than the width, it follows that the length must be greater than 6 m so that the width is positive.

iv. From the graph, an area of 135 m² corresponds to a length of 15 m.

$\therefore$ The dimensions would be 15 m × 9 m.

v.



vi. $\text{Company } A \text{ cost} = \$50\ 000$

$\text{Company } B \text{ cost} = 135 \times 360$

$= \$48\ 600$

$\therefore$ Company B would charge \$1400 less than Company A.

12. Algebra, STD2 A4 2011 HSC 28a

i. $P \propto \dfrac{1}{V}$

$= \dfrac{a}{V}$

♦ *Mean mark (i) 39%*
COMMENT: Expressing the proportional relationship $P \propto \dfrac{1}{V}$ as the equation $P = \dfrac{k}{V}$ is a core skill here.

ii. When $P = 3,\ V = 2$

$3 = \dfrac{a}{2}$

$a = 6$

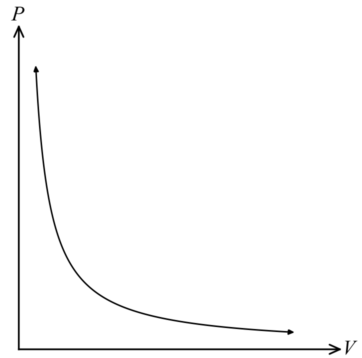
Need to find P when $V = 4$

$P = \dfrac{6}{4}$

$= 1\dfrac{1}{2}$

♦ Mean mark (ii) *47%*

iii.



♦♦ *Mean mark (iii) 26%*
COMMENT: An inverse relationship is reflected by a hyperbola on the graph.

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13. Algebra, STD2 A4 2018 HSC 29c

i. $D \propto \dfrac{1}{A} \Rightarrow D = \dfrac{k}{A}$

♦♦ *Mean mark part (i) 22%.*

When $D = 15, A = 300$

$15 = \dfrac{k}{300}$

$k = 4500$

$\therefore D = \dfrac{4500}{A}$

ii. $4 = \dfrac{4500}{A}$

$\therefore A = \dfrac{4500}{4}$

$= 1125\ \text{cm}^2$

♦♦ *Mean mark part (ii) 33%.*

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14. Algebra, STD2 A4 2022 HSC 24

♦♦ Mean mark part (a) 29%.

a. $M \propto \dfrac{1}{T}$

$$M = \frac{k}{T}$$

$$12 = \frac{k}{15}$$

$$k = 15 \times 12$$

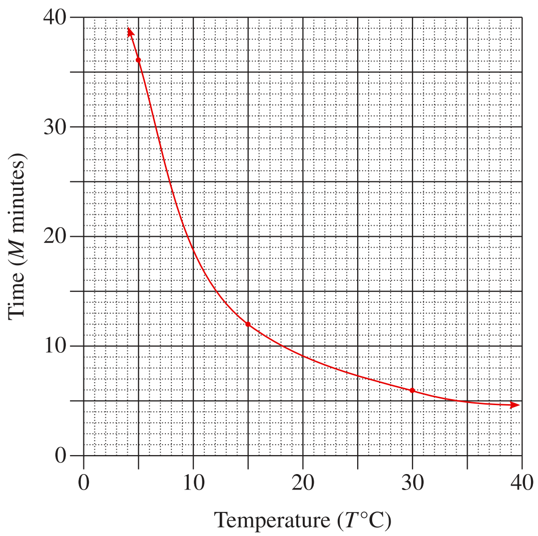
$$= 180$$

$$\therefore M = \frac{180}{T}$$

b.

♦ Mean mark 44%.

T	5	15	30
M	36	12	6



15. Algebra, STD2 A4 SM-Bank 10

i. $t \propto \dfrac{1}{d}$

$$t = \frac{k}{d}$$

$$108 = \frac{k}{5}$$

$$k = 540$$

Find t when $d = 6$:

$$t = \frac{540}{6}$$

$$= 90$$

ii. Find d when $t = 120$:

$$120 = \frac{540}{d}$$

$$d = \frac{540}{120}$$

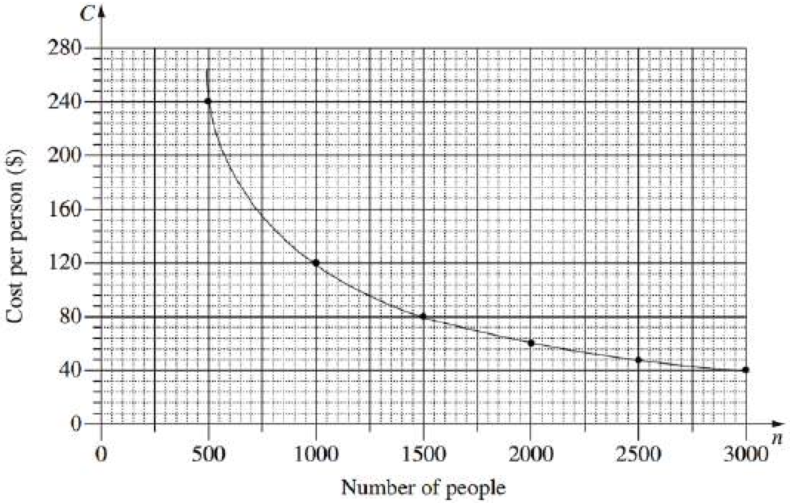
$$= 4.5 \text{ metres}$$

16. Algebra, STD2 A4 2014 HSC 29a

i.

Number of people (n)	500	1000	1500	2000	2500	3000
Cost per person (C)	240	120	80	60	48	40

ii.



iii. $C = \dfrac{120\ 000}{n}$

♦ *Mean mark (iii) 48%*

iv. Limitations can include:

- n must be a whole number
- $C > 0$

♦♦♦ Mean mark (iv) 7%
COMMENT: When asked for limitations of an *equation*, look carefully at potential restrictions with respect to *both the domain and range*.

v. If $C = 94$

$$\Rightarrow 94 = \frac{120\ 000}{n}$$

$$94n = 120\ 000$$

$$n = \frac{120\ 000}{94}$$

$$= 1276.595\ldots$$

♦ *Mean mark (v) 38%*

$\therefore$ Cost cannot be \$94 per person, because n isn't a whole number.

17. Algebra, STD2 A4 2004 HSC 28a

i. $R = \dfrac{w}{h^2}$

When $w = 72$ and $h = 1.5$ m

$$R = \frac{72}{1.5^2}$$

$$= 32$$

ii. Find w if $R = 25$ and $h = 1.6$

$$25 = \frac{w}{1.6^2}$$

$$w = 25 \times 1.6^2$$

$$= 64 \text{ kg}$$

$\therefore$ Weight Fred should lose

$$= 72 - 64$$

$$= 8 \text{ kg}$$